RESEARCH EXPLAINER

Where Does Label Noise Hurt Ensembles?

A 0-1 Bias–Variance Decomposition of Bagging, Random Forests and Boosting under Symmetric, One-Sided and Boundary Label Noise

> **The whole research in one paragraph**
> Real datasets contain **wrong labels**. We train popular ensemble models on data where we deliberately make some labels wrong, in **three different ways**, and measure not just *how much* each model gets worse, but *why*: does it learn a **wrong rule** (bias) or become **unstable** (variance)? The answer tells practitioners which model and which fix to use for their kind of noisy data.

1. Five ideas you need first

Idea	Simple meaning
Decision tree	A flowchart of yes/no questions that ends in a prediction. Simple, but unstable: change a few training rows and the tree changes a lot.
Ensemble	Many models vote together instead of trusting one. Like asking 100 doctors instead of 1.
Bagging / Random Forest	*Parallel* ensembles. Each tree trains on a random resample of the data, then they vote. Voting cancels out individual mistakes.
Boosting (AdaBoost, Gradient Boosting, XGBoost)	*Sequential* ensembles. Trees train one after another, each focusing on the examples the previous ones got wrong.
Label noise	Training examples with the wrong answer attached, e.g. a sick patient recorded as healthy.

2. Bias and variance: the two ways a model fails

Think of a model as an archer shooting at a target many times, each time trained on a slightly different sample of data.

	Bias	Variance
What it looks like	Arrows land together, but **off-centre**	Arrows are **scattered** all over
Meaning	The model learned a **wrong rule** and repeats the same mistake	The model is **unstable**; its answer depends on which data it saw (overfitting)
Typical fix	Clean or correct the data; re-weight examples	Average more models; subsample; stop training early

A **bias–variance decomposition** splits a model's total error into these two parts. We measure it by training the same model about 50 times on different resamples and checking: what is its usual answer (is it right?), and how often does it disagree with itself?

Why it matters: the two problems need *different fixes*. If we only know that accuracy dropped, we don't know which fix to apply.

3. Not all label noise is equal

Earlier studies mostly flipped labels *at random*. Real mistakes are rarely random, so we test three types:

Noise type	How labels go wrong	Real-life example
Symmetric	Random labels flipped, any class	Data-entry slips, accidental clicks
One-sided	Only one class gets flipped (positives → negatives)	Missed diagnoses: sick patients recorded as healthy
Boundary	Mostly the hard, confusing cases near the decision boundary	Annotators disagree on borderline cases

4. The problem we are solving

> **Problem statement (simple version)**
> We know that label noise makes ensembles worse, and that boosting usually suffers more than bagging. But nobody has measured **whether the damage is bias or variance, for each type of noise, across both parallel and sequential ensembles**. Without that, a practitioner cannot tell whether to fix noisy data by *averaging more models* (variance fix) or by *cleaning the labels* (bias fix).

What already exists, and what is missing:

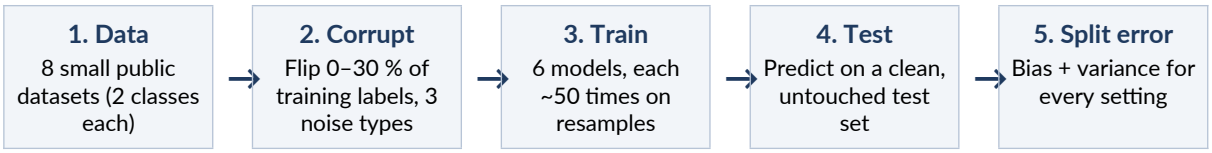
Earlier work	What they showed	What they missed
Dietterich (2000)	Bagging beats boosting under random noise	Only random noise; no bias/variance split
Morampudi et al. (2026)	XGBoost/LightGBM/CatBoost rankings change with noise type	Accuracy only; no bagging/RF; no bias/variance split
He et al. (2026)	Bias–variance theory of noisy ensembles; new algorithms	Theory and new methods, not a measured comparison by noise type
Mohammad (2025)	Discusses ensembles "from a bias –variance view"	No noise added; no real decomposition

Our gap: the first study that *measures* the bias/variance split of noise damage, per noise type, for both bagging-style and boosting-style ensembles.

5. Our four research questions

#	Question (plain words)	Why we ask
RQ1	How much worse does each model get as noise rises from 0 % to 30 %?	Baseline, confirms earlier work
RQ2	Is the extra error **bias or variance**, and does that change with the noise type and model family?	**The core new finding**
RQ3	Is the instability caused by *which* labels were flipped, or by *how the model reacts* to different data samples?	Explains the cause of variance
RQ4	Do common fixes (slower learning, early stopping, smaller samples) repair the damage for every noise type?	Practical advice

6. How we will do it



Models compared: one decision tree (baseline) · Bagging · Random Forest *(parallel)* · AdaBoost · Gradient Boosting · XGBoost *(sequential)*.

Compute: small datasets and fast models, so it all runs on free Kaggle/Colab CPU in a few hours, with no GPU needed.

7. What we expect to find

Expectation	Reason
Random noise → mostly **variance**	Random flips don't move the true rule; they only make the model jittery.
One-sided noise → mostly **bias**	Flipping only one class pushes the learned rule in one direction.
Boundary noise → damage concentrated on **hard cases**	That is exactly where models are already least sure.
Boosting hurt more than bagging	Boosting keeps chasing the wrong labels; voting in bagging outvotes them.
Fixes help **partly**, not fully	Theory (Long & Servedio, 2010) says boosting cannot fully escape noise.

Whether these expectations are confirmed or rejected, the result is useful, because it is measured, not guessed.

8. Why this research matters

- **For practitioners:** a clear guide: *"If your noise looks like X, use model Y and fix Z."*
- **For research:** fills a gap left open by three recent papers (2025–2026).
- **For the course:** covers the whole of Unit 4: ensembles, bagging, boosting, random forests and the bias–variance decomposition.

> **Say it in one sentence**
> "We measure whether label noise hurts ensemble models by making them learn the wrong rule (bias) or by making them unstable (variance), and show that the answer depends on the type of noise and the type of ensemble."